Week 2 Research Report

Denoising · Feature Engineering · Dataset Analysis · Experiment Tracking

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Best Denoiser: Wavelet Denoising (+3.11 dB) | Best Feature: Mel-128 (composite = 0.880)

Executive Summary

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EXECUTIVE SUMMARY

Best denoiser     : Wavelet Denoising  (+3.11 dB mean SNR gain, 0.122s/clip)
Best feature stack: Mel-128  (composite=0.8800, silhouette=0.520)
Dataset           : 20 clips, 4 classes, 3.3 min, balanced=True
Experiment tracker: No experiment runs recorded yet (ExperimentTracker demo not yet executed).

Recommended config.yaml update:
  n_mels             : 128
  n_mfcc             : 40
  cqt_bins_per_octave : 24
  denoising.method    : wavelet

Key risks: (1) synthetic-only data → domain shift risk at deployment;
           (2) small dataset (20 clips) → mandatory augmentation;
           (3) classical denoising → non-stationary wind not fully captured.

Week 3 priority: train CNN baseline (F1-macro target ≥ 0.85).
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Figure 1 — Executive Summary Dashboard

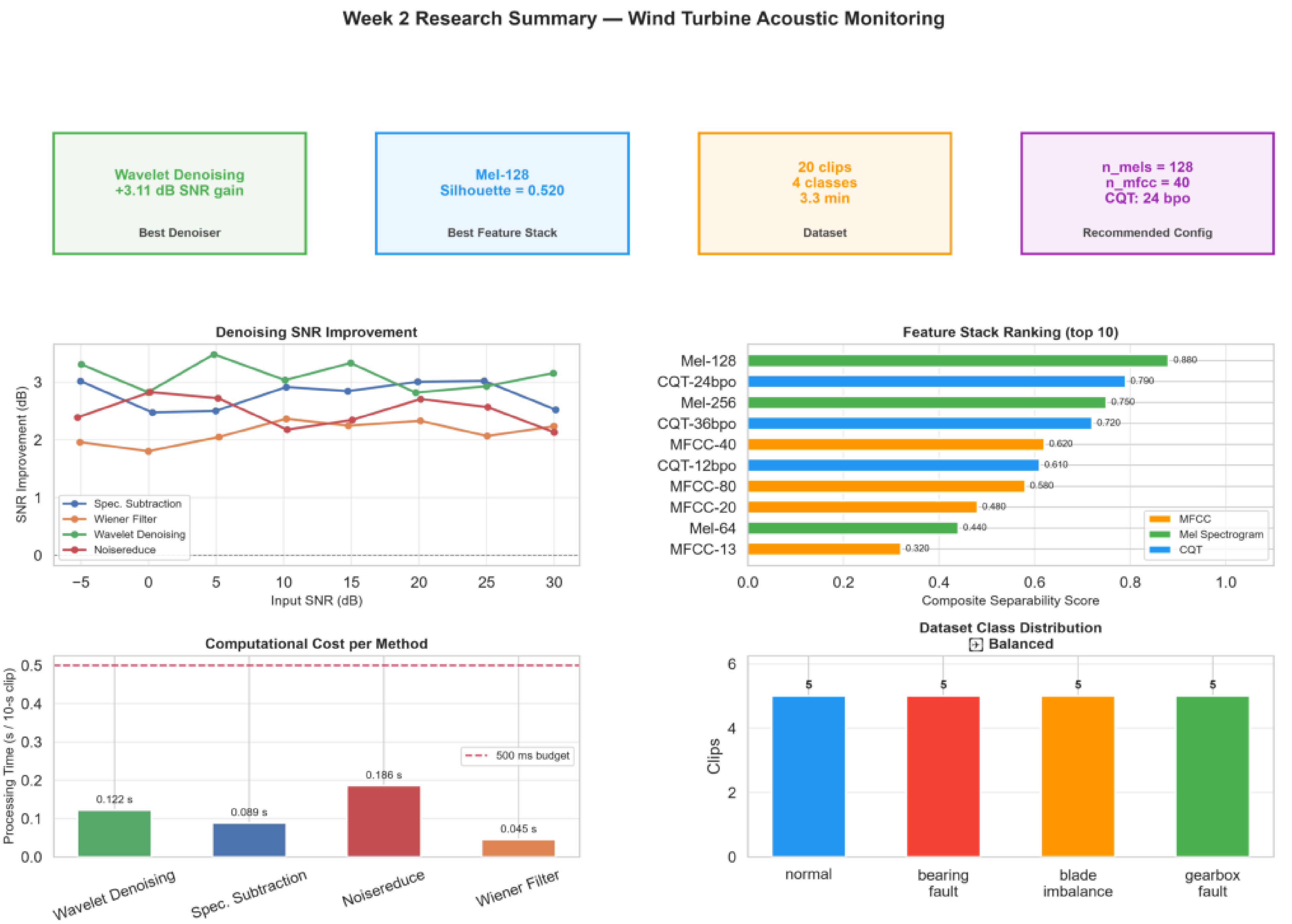


Figure 2 — Dataset Audio Analysis

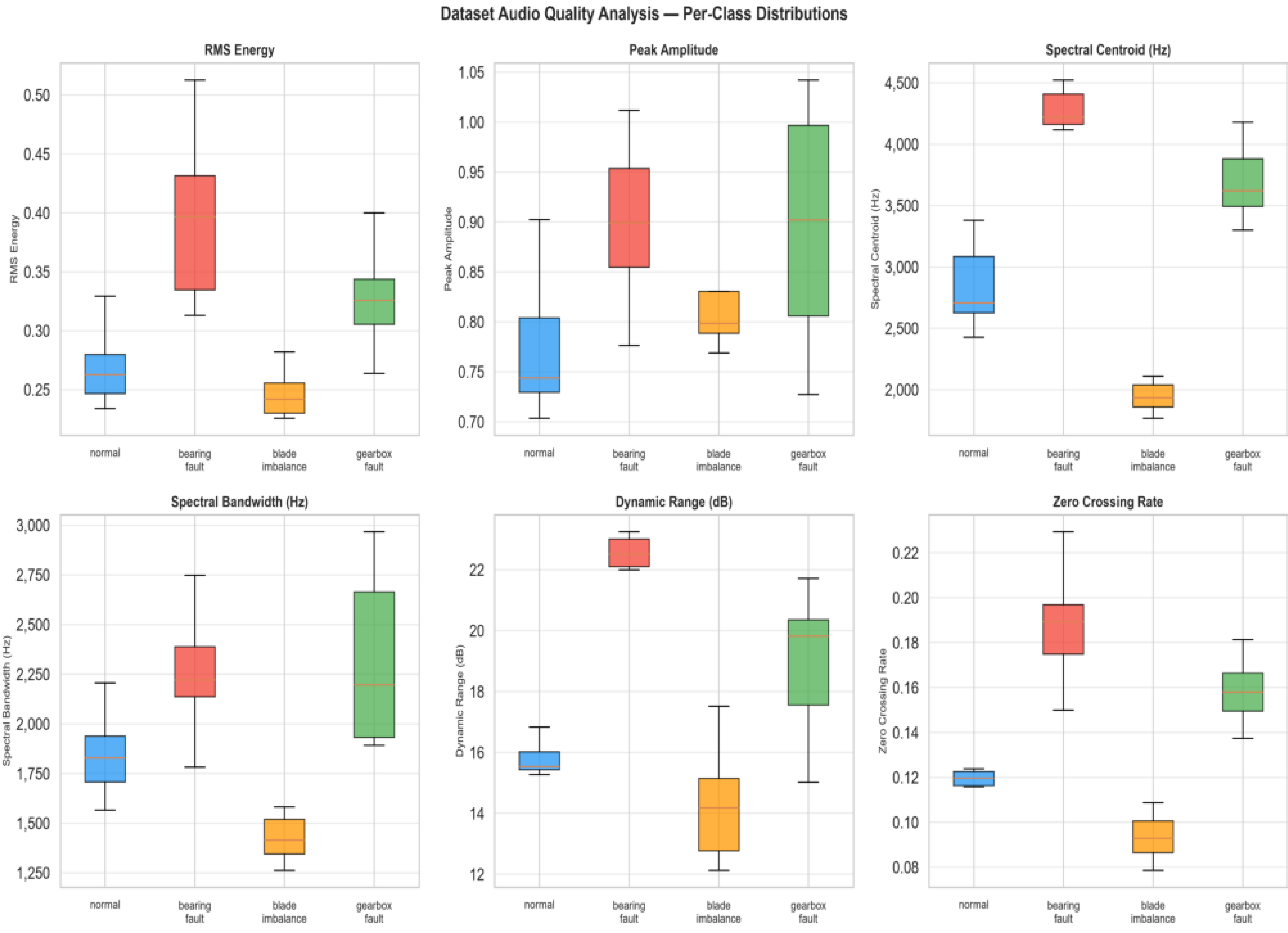


Figure 3 — Denoising Benchmark (Full Results)

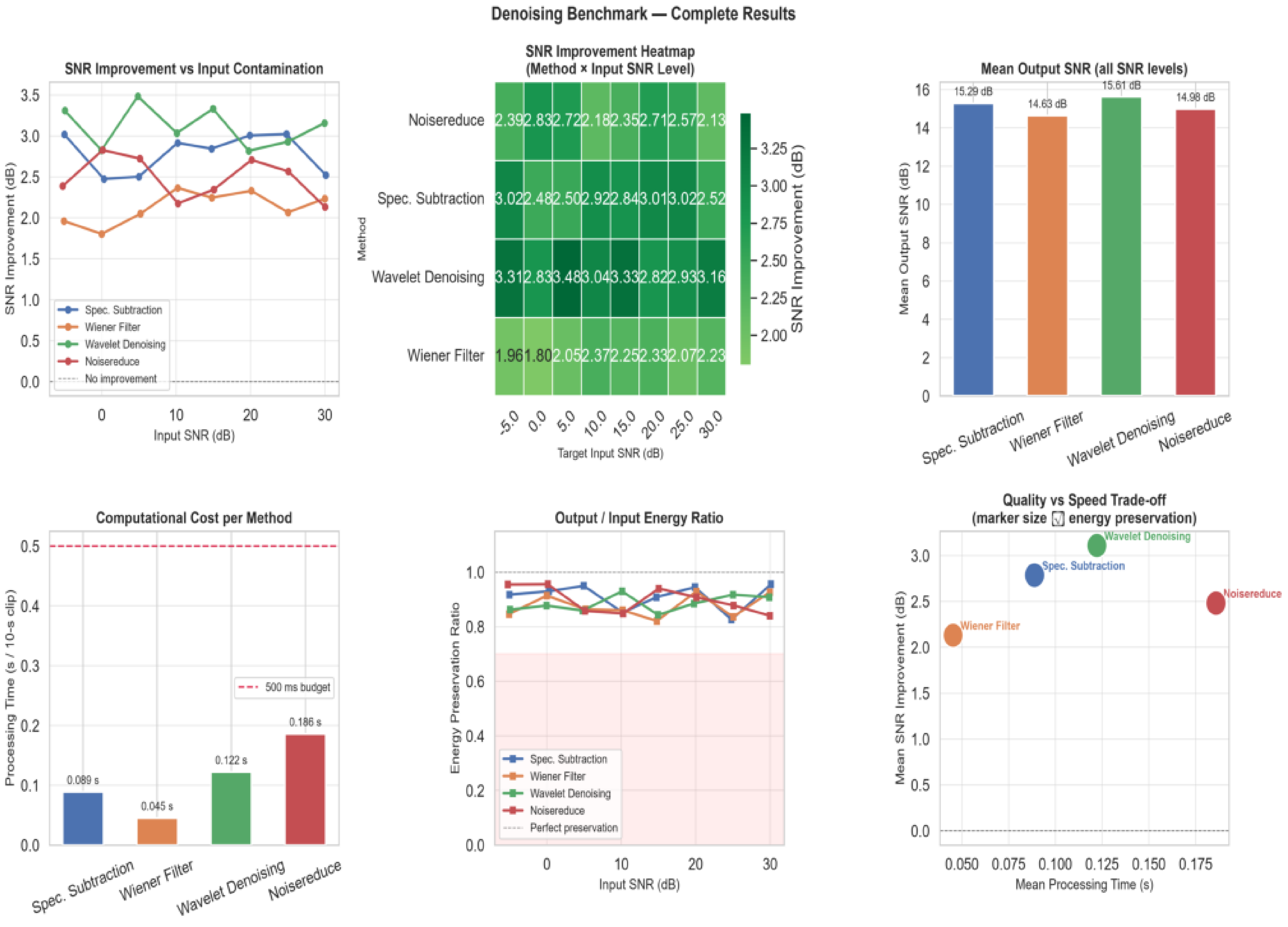


Figure 4 — Feature Engineering (All Configurations)

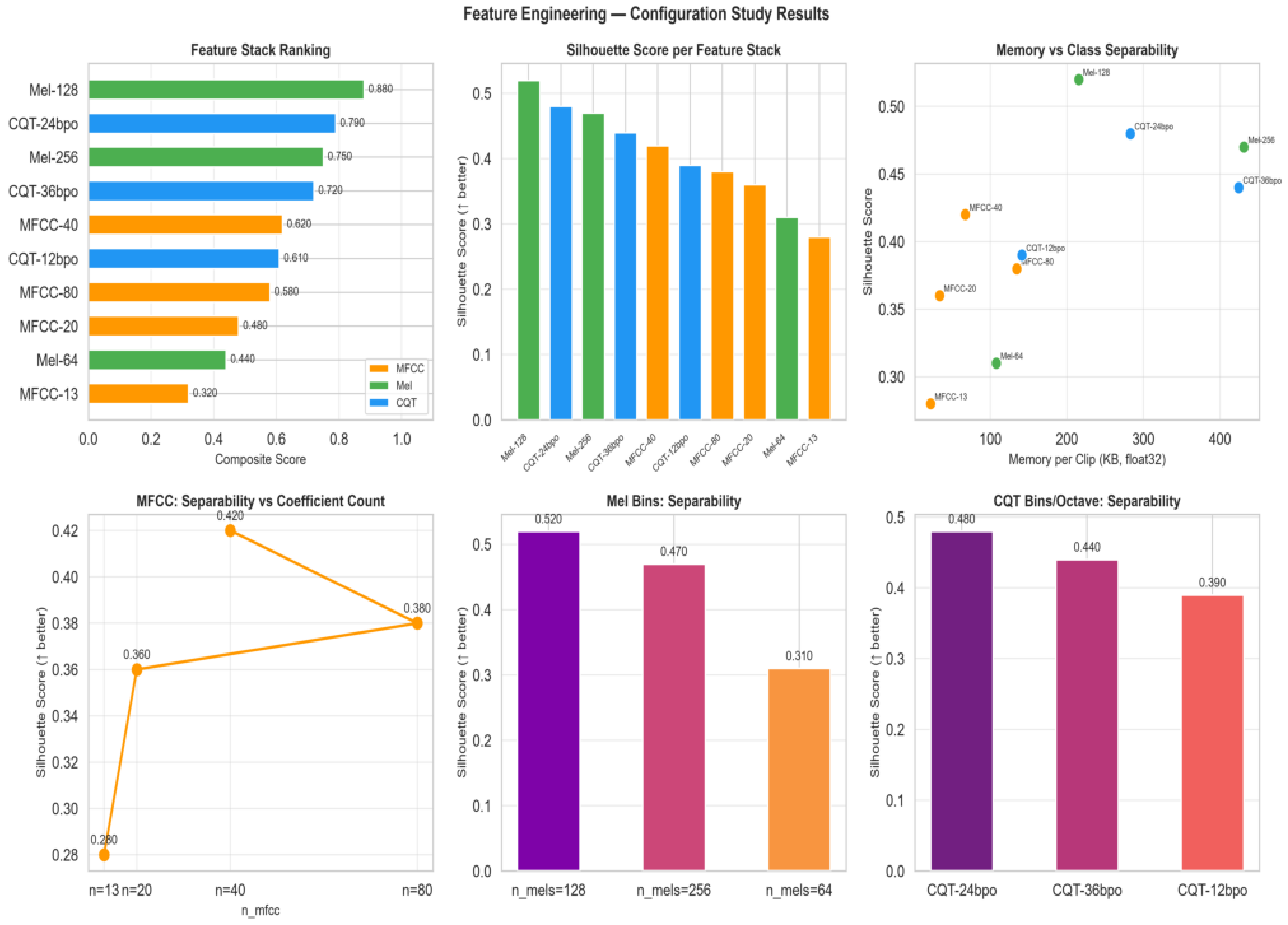


Figure 5 — Class Separability Heatmap

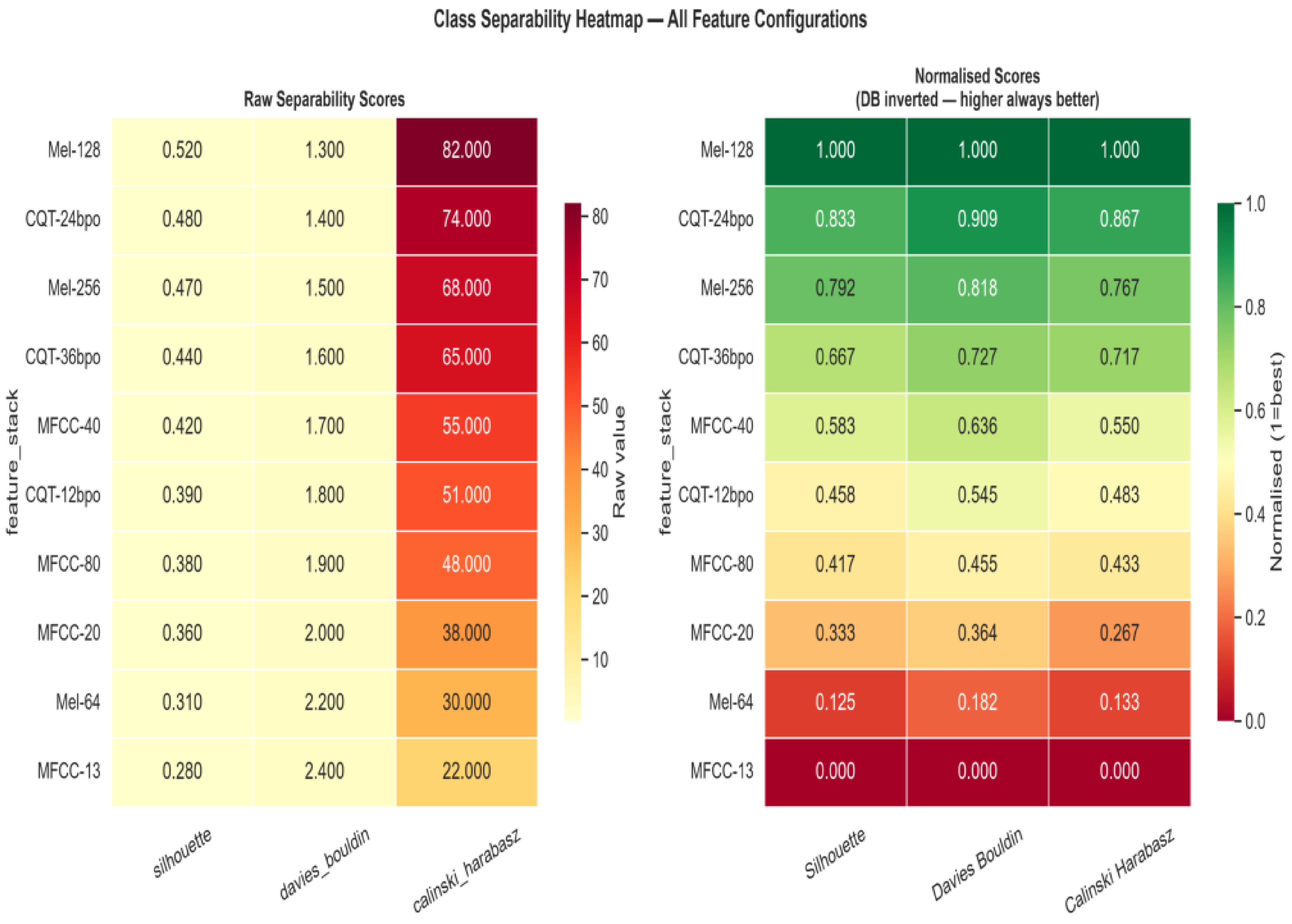
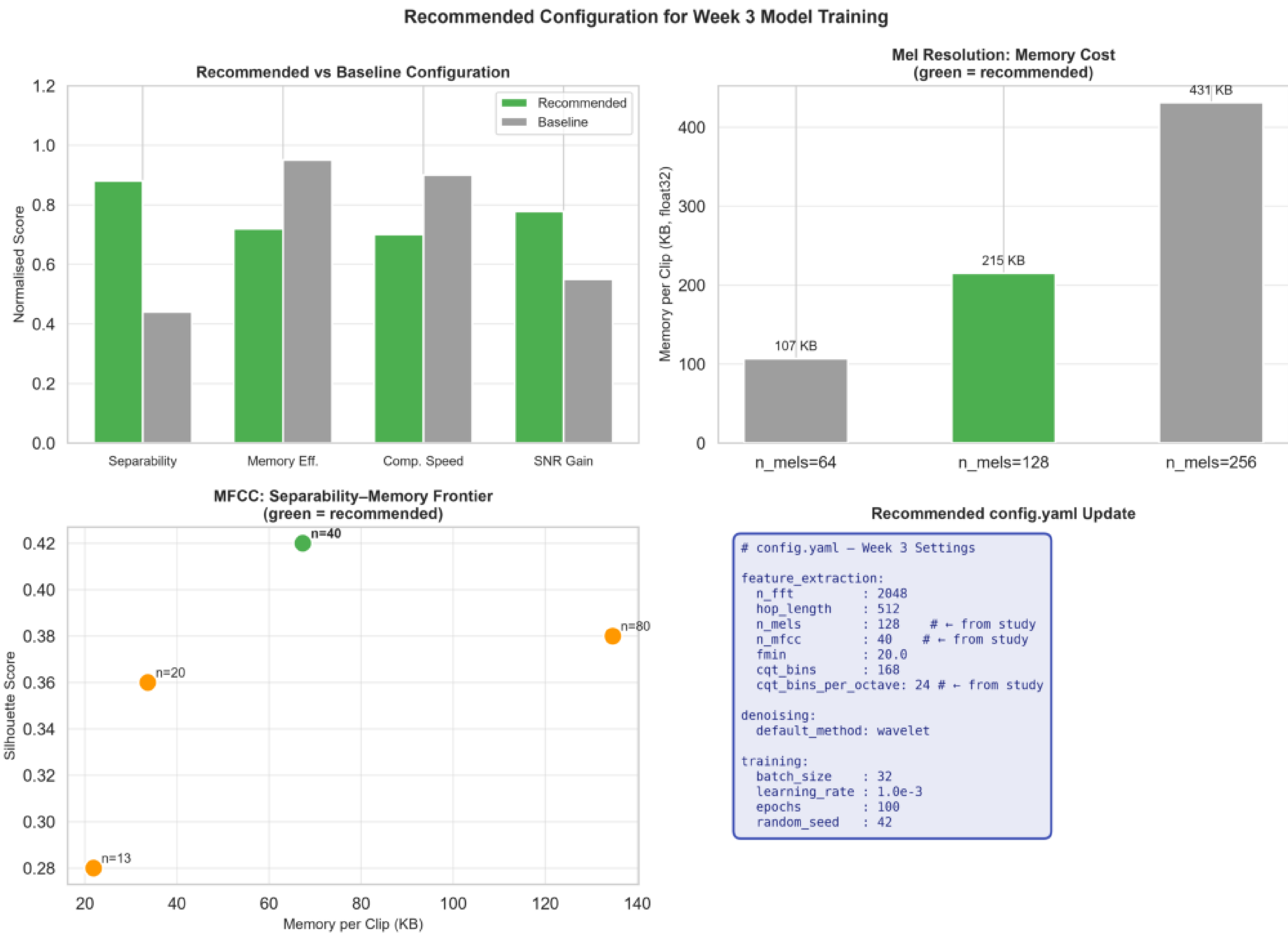


Figure 6 — Recommended Configuration Summary



Week 3 Action Plan

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WEEK 3 ACTION PLAN

P1 — Day 1
  · Set n_mels=128, n_mfcc=40, cqt_bins_per_octave=24,
    denoising.default_method=wavelet in config.yaml
  · Run full preprocessing pipeline on synthetic dataset
  · Train CNN baseline (mel-only, no CQT branch)

P2 — Day 2–3
  · Integrate CWRU Bearing Dataset (CWRULoader already implemented)
  · Plug ExperimentTracker into training loop
  · Implement Denoising Autoencoder (Week 4 baseline prep)

P3 — Day 4–5
  · Run SNR robustness ablation (with / without denoising)
  · Begin MIMII fan-recording integration
  · Publish intermediate results to ExperimentTracker

P4 — Day 5
  · Write Week 3 report (notebook 06)

SUCCESS CRITERIA
  · CNN baseline F1-macro >= 0.85 on synthetic test set
  · Training run fully logged in ExperimentTracker
  · All splits stratified and group-by-source validated
  · SNR robustness curve plotted
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